

L6 LiDAR Perception Quiz

1. A LiDAR pulse returns after 1.33 μ s. Using $R = c \cdot \Delta t / 2$, how far away is the target?

- A) About 200 m
- B) About 400 m
- C) About 20 m
- D) About 2 km

ANS:

2. A return has range $r = 20$ m, azimuth $\theta = 90^\circ$ and elevation $\phi = 0^\circ$. Which Cartesian point (x, y, z) does it map to?

- A) (20, 0, 0)
- B) (0, 20, 0)
- C) (0, 0, 20)
- D) (-20, 0, 0)

ANS:

3. Which statement correctly compares pulsed time-of-flight LiDAR with FMCW LiDAR?

- A) Pulsed ToF measures radial velocity directly, while FMCW cannot measure it at all
- B) FMCW measures the round-trip time of a very short laser pulse like a stopwatch
- C) FMCW measures radial velocity directly via the Doppler shift, but is more complex
- D) FMCW is mass-produced today and is cheaper than a pulsed time-of-flight sensor

ANS:

4. What is the idea of early fusion of camera and LiDAR, e.g. PointPainting?

- A) Run a camera detector and a LiDAR detector separately and merge their boxes
- B) Project the points into the image and append the segmentation scores to each point
- C) Fuse the backbone features of both sensors in one shared BEV feature map
- D) Replace the LiDAR by a monocular depth estimator that works on the images

ANS:

5. A LiDAR has an angular resolution of $\Delta\theta = 0.2^\circ$. What is the approximate spacing between neighbouring points at 100 m ($s = R \cdot \tan \Delta\theta$)?

- A) About 0.035 m
- B) About 3.5 m
- C) About 0.35 m
- D) About 0.17 m

ANS:

6. Why do 1550 nm LiDARs allow a higher pulse energy than 905 nm LiDARs?

- A) Light at 1550 nm is not absorbed by water in the atmosphere at all, so power rises
- B) Light at 1550 nm is absorbed before the retina, so the eye-safety limit is higher
- C) Silicon detectors are more sensitive at 1550 nm than they are at 905 nm
- D) The 1550 nm lasers are cheaper and need much less electrical power to run

ANS:

7. Why does the received LiDAR signal in fog or rain fall as $\exp(-2\alpha R)$?

- A) The detector always averages over two consecutive laser pulses before it reports
- B) The symbol α is defined as twice the extinction coefficient of the air
- C) The light travels out and back, so the attenuation acts over a path of $2R$
- D) Rain reflects exactly half of all the pulses back to the sensor immediately

ANS:

8. Which failure mode creates 'ghost points' beyond walls or vehicles?

- A) Dark targets that return less than 5 % of the emitted photons
- B) Multipath: the pulse bounces over two surfaces before it returns

- C) Retroreflector blooming around road signs and license plates
- D) Motion distortion of a spinning sweep at high vehicle speed

ANS:

9. A 128-channel LiDAR has 1,800 azimuth steps and rotates at 10 Hz. At most how many points does it produce per second?

- A) About 230,000
- B) About 23 million
- C) About 2.3 million
- D) About 18,000

ANS:

10. A 0.5° rotation error in the LiDAR extrinsic calibration shifts a point at 100 m laterally by about how much?

- A) About 0.09 m
- B) About 9 m
- C) About 0.5 m
- D) About 0.9 m

ANS:

11. A spinning LiDAR needs about 100 ms per sweep and the car drives at 20 m/s. What does deskewing do about the resulting 2 m of motion?

- A) It discards all the points that were measured in the first half of the sweep
- B) It increases the rotation speed of the sensor from 10 Hz to 20 Hz
- C) It transforms each point with the vehicle pose at its own timestamp
- D) It applies a single pose, taken at the end of the sweep, to all points

ANS:

12. An $80\text{ m} \times 80\text{ m} \times 4\text{ m}$ region at 0.1 m voxels has 25.6 M voxels, but a sweep has only $\approx 0.19\text{ M}$ points. What does this motivate?

- A) Larger voxels combined with dense 3D convolution over the whole region of interest
- B) Using fewer LiDAR channels so that fewer points are produced in every sweep
- C) Dropping the height coordinate and using only the raw intensity of each point
- D) Sparse convolution (SECOND) or pillars, as dense 3D convolution mostly processes empty space

ANS:

13. RANSAC fits a ground plane (3 points per sample) with inlier ratio $w = 0.5$ and confidence $p = 0.99$. How many iterations $N = \ln(1 - p)/\ln(1 - w^3)$ are needed?

- A) About 35
- B) About 17
- C) About 169
- D) About 4,603

ANS:

14. What is the main advantage of DBSCAN over k-means for clustering LiDAR points?

- A) It always finds exactly one cluster for each object that is in the scene
- B) It is faster than every other clustering algorithm when it is run on a GPU
- C) It uses one single ϵ that works equally well at near and at far ranges
- D) The number of clusters is not needed, and sparse points are rejected as noise

ANS:

15. A parked car is rotated by about 30° . Why is a min-area rectangle a better box than an axis-aligned one?

- A) The axis-aligned box under-estimates the height of the car that is in the scene
- B) The axis-aligned box over-estimates the footprint (15.2 vs. 8.7 m²) and gives no heading
- C) The min-area rectangle can be computed without any prior clustering step at all
- D) An axis-aligned box cannot be computed from a LiDAR point cloud in any way

ANS:

16. Why does ICP need a good initial guess, such as odometry?

- A) Nearest-neighbour matching converges only to a local minimum of the alignment error
- B) ICP does not use any point matching and relies only on image features of the scans
- C) ICP can estimate only a translation and can never estimate a rotation of the scan
- D) The SVD solution is valid only for two scans that are perfectly identical

ANS:

17. Why is PointNet invariant to the order of the input points?

- A) A shared per-point MLP followed by max-pooling, which is a symmetric function
- B) It sorts the points by their range before any of them is processed by the network
- C) It first voxelizes the points into a regular three-dimensional grid of cells
- D) It uses a recurrent network that scans the points of the sweep one by one

ANS:

18. What is the main design idea of PointPillars?

- A) Dense 3D convolution over a fine voxel grid that covers the whole region around the car
- B) Sparse 3D convolution that touches only the non-empty voxels of a regular grid
- C) Raw point processing with hierarchical sampling and grouping of local neighbours
- D) Pillars drop the z split, so a pseudo-image is processed with fast 2D convolutions only

ANS:

19. Which evaluation rule is used for KITTI cars, and how does nuScenes differ?

- A) KITTI: $\text{IoU} \geq 0.7$ counts as correct; nuScenes matches by center distance (0.5 to 4 m)
- B) KITTI matches by center distance, while nuScenes uses an IoU threshold of 0.7 for cars
- C) Both count only the number of points that fall inside the predicted bounding box
- D) Both use classification accuracy without any measure of the localization error

ANS:

20. Why does the lecture stress worst-case rather than average latency, and what does 100 ms mean at a 60 m/s closing speed?

- A) The average latency is always larger than the worst case; 100 ms is 0.6 m
- B) Only the throughput of the system matters for safety; 100 ms is 60 m
- C) Latency is irrelevant at a 10 Hz sweep rate; 100 ms corresponds to 1 m
- D) One late result can be unsafe, so the maximum matters; 100 ms at 60 m/s is 6 m

ANS: